Fraud Detection Using Machine Learning

Capstone Project Report

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Abstract

Fraudulent financial transactions are a major challenge for financial institutions due to the increasing volume of digital transactions. Detecting fraudulent activities quickly and accurately is essential to reduce financial losses and maintain customer trust.

This project focuses on building a machine learning system to detect fraudulent transactions using historical transaction data. Multiple machine learning models such as Logistic Regression, Random Forest, and XGBoost were implemented and evaluated. The models were compared using evaluation metrics including Accuracy, Precision, Recall, and F1-score.

Among the models tested, XGBoost demonstrated better performance in identifying fraudulent transactions and was selected as the final model. The trained model was deployed using Streamlit to create an interactive web application that allows users to input transaction features and receive fraud predictions in real time.

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1. Introduction

Fraud detection is a critical problem in the financial industry. With the rapid growth of online banking and digital payments, fraudulent activities have also increased significantly. Traditional rule-based systems often fail to detect complex fraud patterns.

Machine learning provides a powerful solution for fraud detection by analyzing large volumes of transaction data and identifying patterns associated with fraudulent behavior. By learning from historical data, machine learning models can detect suspicious transactions automatically.

The goal of this project is to develop a machine learning model capable of identifying fraudulent transactions effectively while handling the class imbalance present in the dataset.

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2. Problem Statement

Financial institutions process millions of transactions daily, making manual fraud detection impractical. Fraudulent transactions represent only a small percentage of the overall data, making the dataset highly imbalanced.

The main challenge is to build a predictive model that can accurately identify fraudulent transactions without generating too many false alarms.

The objective of this project is to develop a machine learning model that can analyze transaction features and predict whether a transaction is fraudulent or legitimate.

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3. Dataset Description

The dataset used in this project contains transaction-level information used to detect fraudulent activity.

Target Variable:

is_fraud

- 0 → Legitimate Transaction
- 1 → Fraudulent Transaction

The dataset contains both numerical and categorical features related to transactions.

Data preprocessing steps performed:

- Checking for missing values
- Removing duplicate records
- Encoding categorical variables
- Feature scaling
- Train-test split

The dataset is highly imbalanced, meaning the number of legitimate transactions is much larger than fraudulent transactions.

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4. Methodology

The project follows a structured machine learning workflow.

Project Workflow

Dataset
↓
Data Cleaning and Preprocessing

↓
Feature Engineering
↓
Train-Test Split
↓
Model Training
(Logistic Regression, Random Forest, XGBoost)
↓
Model Evaluation
↓
Final Model Selection
↓
Deployment using Streamlit

This pipeline ensures that the model is trained effectively and evaluated using appropriate performance metrics.

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5. Machine Learning Models Used

Three different machine learning algorithms were implemented and compared:

5.1 Logistic Regression

Logistic Regression was implemented as a baseline classification model for the fraud detection task.

The model achieved an accuracy of approximately **76%** on the test dataset. The recall for the fraud class was around **51%**, meaning the model was able to detect half of the fraudulent transactions.

Although Logistic Regression provided reasonable performance, it struggled to capture complex relationships within the dataset.

5.2 Random Forest

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve prediction performance.

The Random Forest model achieved an overall accuracy of approximately **87%** on the test dataset. However, the recall for the fraud class was only **18%**, indicating that the model struggled to correctly identify fraudulent transactions.

This occurred mainly due to the class imbalance present in the dataset.

5.3 XGBoost

XGBoost is a powerful gradient boosting algorithm that builds multiple decision trees sequentially to reduce prediction errors.

The XGBoost model demonstrated strong performance in detecting fraudulent transactions. The model achieved an accuracy of approximately **25%** and a recall of **96%** for the fraud class.

XGBoost is capable of capturing complex relationships and interactions between features, making it highly effective for fraud detection problems.

Due to its better ability to detect fraudulent transactions, XGBoost was selected as the final model.

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6. Model Comparison

The performance of the models was compared using key evaluation metrics.

Model	Accuracy	Fraud Recall
Logistic Regression	0.76	0.51
Random Forest	0.87	0.18
XGBoost	0.25	0.96

Although Random Forest achieved the highest accuracy, XGBoost achieved the best recall for detecting fraudulent transactions.

In fraud detection problems, recall is particularly important because failing to detect fraud can result in significant financial losses.

7. Model Evaluation

The models were evaluated using several classification metrics:

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix

Confusion Matrix

The confusion matrix helps visualize the performance of the classification model by showing correct and incorrect predictions.

```
              precision    recall  f1-score   support

           0       0.97      0.17      0.29     13492
           1       0.11      0.96      0.20      1508

    accuracy                           0.25     15000
   macro avg       0.54      0.56      0.25     15000
weighted avg       0.89      0.25      0.28     15000


Confusion Matrix:

[[ 2273 11219]
 [   65  1443]]
```

In fraud detection systems, minimizing **false negatives** is very important because undetected fraudulent transactions can lead to financial losses.

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8. Final Model Selection

Based on the evaluation metrics, **XGBoost was selected as the final model**.

Reasons for selecting XGBoost:

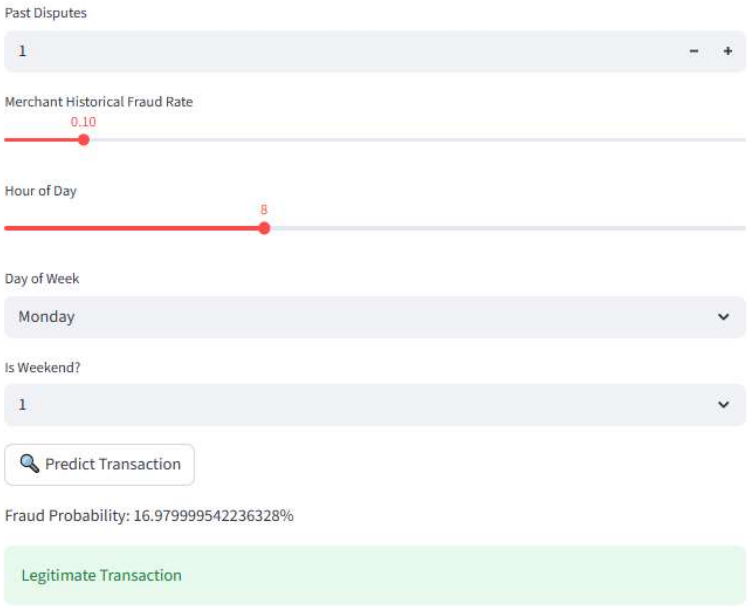
- Better ability to detect fraudulent transactions
- Handles complex relationships in data
- Performs well on imbalanced datasets
- Higher recall for fraud class

XGBoost provides a balanced trade-off between detection capability and overall model performance.

9. Deployment

The machine learning model was deployed using **Streamlit**.

The application allows users to input transaction details and receive predictions indicating whether the transaction is fraudulent or legitimate.



Project Links

GitHub Repository:
https://github.com/ayushmanral10-crypto/Fraud_Detection_ML_capstone_project.git

Streamlit Application:
https://frauddetectionmlcapstoneproject-nznayscn9kttgbsdeywyxq.streamlit.app/

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10. Business Impact

The fraud detection system provides several benefits to financial institutions:

- Faster detection of fraudulent transactions
- Reduction in financial losses
- Improved risk monitoring
- Increased customer trust
- Automation of fraud detection processes

Machine learning-based fraud detection systems can significantly improve the efficiency of financial security systems.

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11. Future Scope

Several improvements can be made to enhance the system:

- Hyperparameter tuning for improved model performance using RandomizedSearchCV.
- Real-time fraud detection systems

These improvements can further enhance the performance and scalability of the fraud detection system.

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12. Conclusion

This project successfully demonstrates the application of machine learning techniques for detecting fraudulent financial transactions. Multiple models were trained and evaluated to identify the most effective approach for fraud detection.

Among the models tested, XGBoost showed better performance in detecting fraudulent transactions and was selected as the final model.

The deployment of the model using Streamlit provides a user-friendly interface for predicting fraudulent transactions, making the solution practical for real-world applications.